

COURSE SYLLABUS

CS365: Introduction to Computer Organization and Architecture

COURSE DESCRIPTION

Credit points	3 credits (45 hrs)	
Level	Undergraduate	
Instructor	Ngo Lam Trung, PhD Department of Computer Engineering, School of Information and Communication Technology Hanoi University of Science and Technology	
Teaching time	19/2/2024 – 8/6/2024	
Location	Hanoi University of Science and Technology	
Prerequisites	CS310: Foundation of Computer Science	
Recommended background knowledge	Digital Logic, C programming language	
Subject description	This course is an introduction to computer architecture for students in IT-related fields. In this course, students will study: (1) Computer organization: the arrangement of computer components (CPU, memory, peripherals (I/O) and system bus) and their relationship. (2) The instruction set architecture (ISA). (3) The CPU structure (ALU, registers, decoder), pipelining technique. (4) Memory and I/O system.	
Objectives & Out-come	- Basic knowledge of the architecture and operation inside a computer. - Understanding of the relationship between hardware and software, and the basic laws for computer performance evaluation. - Understanding of MIPS instruction set architecture, and the ability to write MIPS programs to solve computing problems. - Ability to understand and analyse the design concept of microprocessor for modern computer architecture. - Team work and communication skill to discuss and solve assignments	
Assessment/ Evaluation	Attendance/Attitude	10%
	Assignment(s)	10%
	Group presentation	10%
	Mid-term exam	30%
	Final exam	40%
Prescribed Textbook(s)	Computer Organization and Design, 4th Edition, Patterson & Hennessy, MK Pub, 2008.	

COURSE CONTENTS & SCHEDULE

Class No.	Contents	No. of Hours			Ref./Resources	Assignment(s)
		Lect.	Exr.	Prc.		
1	Course Introduction and Orientation Chapter 1: Introduction 1.1 Computer Abstraction and Technology - Computer history - Computer anatomy - Computer program - Hardware and software 1.2 Performance Evaluation and Benchmarking - Performance measurement - Enhancing performance (hardware, software...)	3			Chapter 1 [Textbook]	
2	Chapter 2: Instruction Set Architecture 2.1 Overview 2.2 MIPS operands 2.3 MIPS instruction set	3			Chapter 2 [Textbook]	
3	2.3 MIPS instruction set (cont.) 2.4 Basic programming patterns with MIPS instruction set 2.5 Procedures	3			Chapter 2 [Textbook]	
4	Exercise and in-class assignment		3			Assignment 1
5	Chapter 3: Computer Arithmetic 3.1 Integer arithmetic 3.2 Floating point arithmetic	3			Chapter 3 [Textbook]	
6	Chapter 4: CPU 4.1 Introduction 4.2 Simple CPU implementation	3			Chapter 4 [Textbook]	
7	4.3 Enhancing performance with pipelining - Overview of pipelining - MIPS pipeline - Pipeline performance - Hazards				Chapter 4 [Textbook]	
8	Assignment 1 submission Practice and in-class assignment		3			Assignment 2
9	Mid-term exam Exam solution		3			

10	Chapter 5: Memory 5.1 Memory hierarchy 5.2 Cache - Overview of cache - Operation - Cache performance	3			Chapter 5 [Textbook]	
11	5.2 Cache (cont.) - Improving cache performance 5.3 Virtual memory 5.4 Virtual machine	3			Chapter 5 [Textbook]	
12	Chapter 6: I/O System 6.1 Introduction to I/O system 6.2 Storage 6.3 Interfacing with I/O system 6.4 I/O performance	3			Chapter 6 [Textbook]	
13	Chapter 7: Multicore and multiprocessors 7.1 Introduction 7.2 Shared memory multiprocessors 7.3 Introduction to GPU Assignment 2 submission Practice and in-class assignment	3			Chapter 7 [Textbook]	Assignment 3: Group presentation
14	Group presentation		3			
15	Final exam Exam solution		3			

Notes:

- Abbreviation: *Lect.* (lecture), *Exr.* (Exercise), *Prc.* (Practise).

- Exercises may include assignment, reports, student's presentation, homework, class exercises... for each class sessions.

- Practical mostly refer to Lab-work or outside practice such as field trip.

- Assignments may include assignments, practical work, reports, exercises ...for each class sessions

Reference Literature:

[1]. Computer Organization and Design, 4th Edition, Patterson & Hennessy, MK Pub, 2008.
[2]. Computer Architecture: From Microprocessors to Supercomputers, Behrooz Parhami, Oxford Univ. Press, New York, 2005.
[3]. Computer Architecture and Organization, 7th Edition, William Stallings, Prentice Hall International, 2006.